MCQs U3

1

If the mean and variance of a binomial random variable are 11.25 and 2.8125, respectively, find the number of trials.

(a) 20 (b) 10 (c) 45 (d) 15

2

The mean and variance of a Binomial random variable are 2 and 1.2, respectively, find $P(X = 0)$.

(a) 0.0778 (b) 0.0102 (c) 0.778 (d) 0.25

3

For Bernoulli distribution with probability p of a success and q of a failure, the relation between mean and variance that hold is

(a) Mean $<$ Variance (b) Mean $>$ Variance
(c) Mean $=$ Variance (d) Mean $\leq$ Variance

4

.Select the correct option regarding mean and variance of Poisson distribution.

(A) Mean is greater than variance.

(B) Mean is less than variance.

(C) Mean is equal to variance.

(D) Mean and variance both are equal to 1.

5

.Select the correct option regarding mean and variance of Negative Binomial distribution.

(A) Mean is greater than variance.

(B) Mean is less than variance.

(C) Mean is equal to variance.

(D) Mean and variance both are equal to 1.

6.

.If a company producing the large number of items, then the probability of 4 defective items can be obtained by

(A) Bernoulli distribution

(B) Binomial distribution

(C) Negative Binomial distribution

(D) Poisson distribution

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For the binomial distribution, n=5. If P(X=4)=P(X=3) then p=.....

a)1/3 b)3/4 c)1 d)2/3

8.

Mean and variance of binomial distribution is 4 and 3. Find P(X≥2)

a) 0.053454
b) 0.010023
c) 0.936524
d) 0.9993

9.

A basketball player makes free throws with a 70% success rate. What's the probability that their fifth successful free throw happens on their 8th attempt?

(a) 0.159 (b) 0.354 (c) 0.581 (d) 0.716

10

If X~NBD(r=2,p=1/2), then variance of X is

(a) 4 (b) 8
(c) 2 (d) 1

11

If the probability is 0.4 that a child exposed to a certain disease will contract it, what is the probability that the 10th child expose to the disease will be the third to catch it?

a) 0.1852
b) 0.0645
c) 0.0548
d) 0.645

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If the probability of success in a single trial is p, what is the probability that the first success occurs on the k-th trial in a geometric distribution?

A) $p(1-p)^{k-1}$
B) $(1-p)^k p$
C) $(1-p)^{k-1}(1-p)$
D) $(1-p)^k$

13

If X~Geom(1/2), then mean of X is

(a) 1 (b) 2
(c) 1/2 (d) 4

14

For Poisson distribution find P(2) given λ =0.7[$e^{-0.7}$ =0.497]

a) 0.13 b) 0.14 c) 0.12 d) 0.9

15

If X~P(5), then the ratio of mean to the variance is

(a) 1 (b) 5
(c) 1/5 (d) 25

16

Five coins are tossed 3200 times. What is the probability of getting 5 heads two times

a) e^(-100) × 5000
b) e^(-100) × 10000
c) e^(-100)
d) None of the above

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The variance of a random variable X using MGF is given by

A. $M_X''(0) - [M_X'(0)]^2$
B. $M_X''(1) - [M_X'(1)]^2$
C. $M_X'(0)$
D. None of the above

18

If MGF of X is $M_X(t) = e^{4t+9t^2}$ then mean of X is

(a) 4 (b) 8 (c) 9 (d) (13)

19

Which random variable does the MGF $M(t) = e^{\lambda(e^t-1)}$ correspond to-

(a) Binomial Distribution (b) Normal Distribution
(c) Poisson Distribution (d) None of these

20

What is the moment-generating function (MGF) of the geometric distribution?

a) $M_X(t) = \binom{k+r-1}{k} \cdot p^r \cdot q^k$
b) $M_X(t) = \left(\frac{p}{1-qe^t}\right)^r$
c) $M_X(t) = (1-p)^{k-1} \cdot p$
d) $M_X(t) = \frac{pe^t}{1-(1-p)e^t}$

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10. The moment generating function of r.v. X can be obtained from

(A) Expectation of X

(B) Variance of X

(C) Expectation of tx

(D) Expectation of exponential (tX)

22

If a random variable X has a Moment generating function $M(t)=\frac{e^t}{3-2e^t}$, then mean of X is given by

(a) 2 (b) -2 (c) 3 (d) None of these

23

Which of the following expressions represents the moment-generating function (MGF) of the negative binomial distribution?

a) $M_X(t) = (1-p)^{k-1} \cdot p$
b) $M_X(t) = \frac{pe^t}{1-(1-p)e^t}$
c) $M_X(t) = \left(\frac{p\,e^t}{1-qe^t}\right)^r$
d) $M_X(t) = \left(\frac{qe^t}{1-pe^t}\right)^r$

24

If x is Poisson variate such that P(x=1) =2P(x=2).Then λ and σ are

a) 1 and 1 b) 1 and 2 c) 4 and 2 d) 2 and 1

For a random variable X, m.g.f $M_X(t) = \frac{1}{(1-3t)^2}$ and $Y = 3X$, then $M_Y(t) =$ **[1]**

(A) $\frac{3}{(1-3t)^2}$ (B) $\frac{9}{(1-3t)^2}$ (C) $\frac{1}{(3-t)^2}$ (D) $\frac{1}{(1-9t)^2}$

25

If the mgf of the negative binomial distribution is $\left(\frac{2e^t}{3-e^t}\right)^3$, then the variance is

a) 9/2 b) 9/4 c) 3/2 d) none

26

The m.g.f. of a random variable X is given by $M_X(t) = e^{3(e^t-1)}$. The value of $P(X=1)$.

a) 0.1494 b) 0.1532 c) 0.5671 d) none of these

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If the mgf of the geometric distribution is $\frac{2e^t}{3-e^t}$, then the probability of 1st success in 4th trial is

a) 2/9 b) 1/81 c) 2/81 d) none

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The moment generating function of a random variable X is given by

$$M_X(t) = \frac{1}{6} + \frac{1}{3}e^t + \frac{1}{3}e^{2t} + \frac{1}{6}e^{3t}, \quad -\infty < t < \infty.$$

Then $P(X \leq 2)$ equals

a) 1/3 b) 1/6 c) ½ d) 5/6

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For a random variable X, with m.g.f $M_X(t) = e^{t^2-3t}$ and $E(X^2) = 11$, the value of Variance of X is

a) 20 b) 14 c) 2 d) 9

30

If $X \sim P(x; \lambda)$ with m.g.f. given as $e^{5(e^t-1)}$, then $\mu_2 =$

a) 16 b) 50 c) 30 d) 5